Quiz 1

ECON 300: Labour Economics

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Multiple Choice ($5 \times 1 = 5$ points)

Choose the best answer for each question.

1. Which of the following individuals is classified as **unemployed** in official labour force statistics?
   (a) A full-time student not looking for work
   (b) A retired worker
   (c) A worker on temporary layoff expecting recall
   (d) A discouraged worker who stopped searching

2. The labour force participation rate (LFPR) measures:
   (a) The fraction of the population that is employed
   (b) The fraction of the working-age population that is employed or unemployed
   (c) The fraction of workers who are unemployed
   (d) The fraction of workers actively searching for jobs

3. In the labour–leisure model, an increase in the wage rate makes leisure:
   (a) Cheaper
   (b) More valuable
   (c) More expensive in terms of foregone income
   (d) Irrelevant for labour supply

4. The substitution effect of a wage increase tends to:
   (a) Reduce hours worked
   (b) Increase hours worked
   (c) Leave hours unchanged
   (d) Always dominate the income effect

5. Which of the following would most likely **increase** a worker's reservation wage?
   (a) A fall in non-labour income
   (b) Lower commuting costs
   (c) Stronger preference for leisure
   (d) A higher market wage

Problem 1: Labour Force Statistics and Interpretation (7.5 points)

A city has a working-age population (15+) of 1,000,000 people. The breakdown is:

- Employed: 640,000
- Unemployed: 60,000
- Not in the labour force: 300,000

(a) (2.5 points) Calculate the **labour force** and the **labour force participation rate**.

(b) (2.5 points) Calculate the **unemployment rate** and the **employment-to-population ratio**.

(c) (2.5 points) Suppose a recession causes:

- 20,000 employed workers to lose their jobs, and
- 15,000 discouraged workers to stop searching for work.

Recalculate the unemployment rate and explain why it may rise by **less than expected**.

Problem 2: Labour Supply, Market Equilibrium, and Minimum Wage (7.5 points)

A worker's weekly budget constraint is:

$$C = 600 + 25h$$

and utility is:

$$U = C - 0.5h^2$$

The market labour demand and supply are:

$$L^D = 120 - 2w, \qquad L^S = 20 + 3w$$

(a) (2.5 points) Find the worker's **optimal hours worked**.

(b) (2.5 points) Find the **equilibrium wage and employment** in the labour market.

(c) (2.5 points) A minimum wage of $w = 20$ is introduced. Determine whether it is binding and calculate the resulting **unemployment**.

End of Quiz